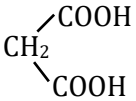
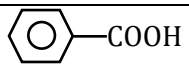
RACE
Regular Analysis Through Continuous Exercise

Carboxylic Acids and its Derivatives : RACE-01

1. A compound of general formula $C_nH_{2n}O_2$ could be

(A) An acid (B) A diketone (C) An ether (D) An aldehyde

2. Match the column :

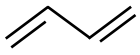
Column-I		Column-II	
(A)	HCOOH	(P)	Mono carboxylic acid
(B)		(Q)	Aromatic acid
(C)		(R)	Di–Carboxylic acid
(D)		(S)	Found in red ants

(A) (A) → R,S; (B) → S; (C) → Q; (D) → P,R (B) (A) → P,Q; (B) → P; (C) → R; (D) → P,S

(C) (A) → P,S; (B) →R; (C) → R; (D) → P,Q (D) (A) → P,R; (B) → R; (C) → S; (D) → P,

3. $CH_3Cl + KCN \longrightarrow$ 'X' $\xrightarrow{\text{Hydrolysis}}$ 'Y' $\xrightarrow{P_2O_5}$ 'Z' . In the above series of reaction 'Z' is

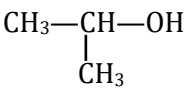
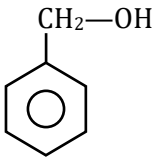
(A) CH_3CN (B) CH_3COOH (C) $(CH_3CO)_2O$ (D) $CH_3—CH_2—OH$

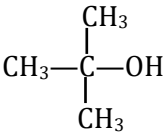
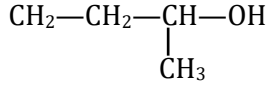
4.  $\xrightarrow[\text{HBr}]{1\text{ mol}}$ (B) $\xrightarrow[(iii)H_3O^{\oplus}]{(i)Mg,\ (ii)CO_2}$ (C) compound C is.

(A) α, β, unsaturated acid (B) β,γ–unsaturated acid

(C) But–2–ene (D) But–1–ene

5. Which of the following alcohol on oxidation give carboxylic acid

(A)  (B) 

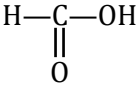
(C)  (D) 

6. Hydrolysis of ester is irreversible in which medium ?

(A) Acidic medium (B) Basic medium

(C) Both acid and Basic medium (D) None of these

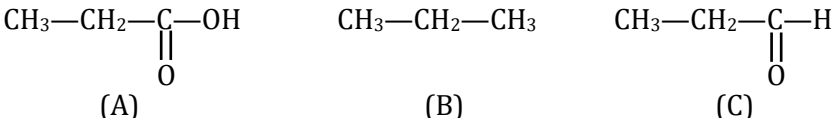
7. Hybridization of C-atom in given compound.



(A) sp^2 (B) sp^3 (C) sp (D) dsp^2

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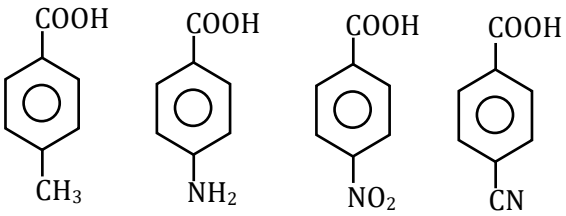
8. Arrange the following compound in order of Boiling point



(A) (B) (C)

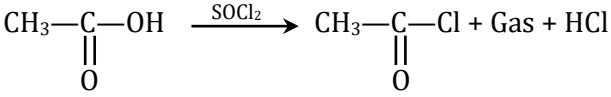
(A) B > A > C (B) B > C > A (C) A > B > C (D) A > C > B

9. Select the correct order of acidic nature of



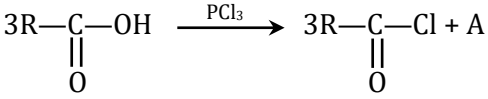
(I) (II) (III) (IV)

(A) III > IV > I > II (B) IV > III > II > I (C) II > I > IV > III (D) I > II > III > IV

10. 

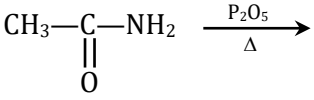
How many total number of lone pair are present in gas A ?

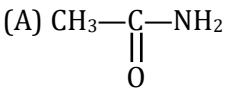
(A) 3 (B) 1 (C) 4 (D) 5

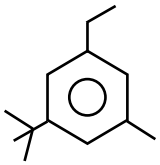
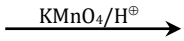
11. 

What is chemical formula of by product A

(A) H_3PO_2 (B) H_3PO_3 (C) H_3PO_4 (D) $POCl_3$

12. 

(A)  (B) $CH_3-C\equiv N$ (C)  (D) 

13.   (P). Number of COOH group present in product "P" is

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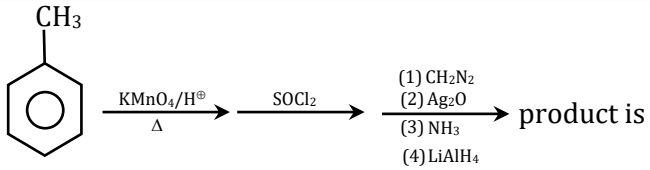
ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	C	C	B	B	B	A	D	A	D
Question	11	12	13							
Answer	B	B	2							

For More Join: @IITJEE_Advanced

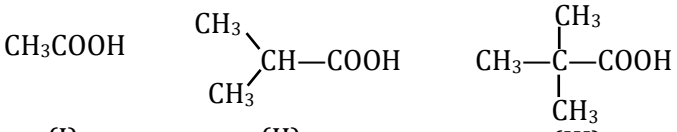
RACE
Regular Analysis Through Continuous Exercise

Carboxylic Acids and its Derivatives : RACE-02

1. 

product is

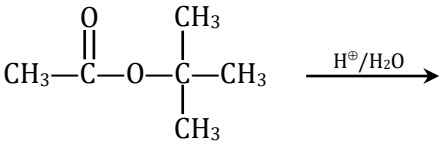
(A) 1° Amine (B) 2° Amine (C) 3° Amine (D) Acid amine

2. 

(I) (II) (III)

Arrange the following acid in order of easy esterification with CH_3OH.

(A) I > II > III (B) III > II > I (C) II >I >III (D) I > III > II

3. 

reverse of esterification follows.

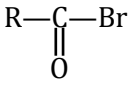
(A) S_N1 (B) S_N2 (C) S_Ni (D) None of these

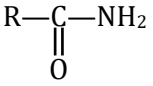
4. Which of the following forms cyclic anhydride

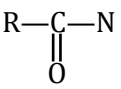
(A) Benzoic acid (B) Succinic acid

(C) Phthalic acid (D) Acetic acid

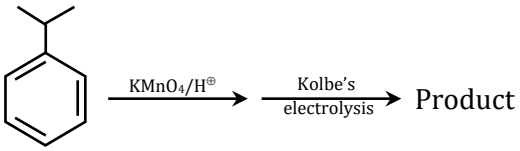
5. Which of the following compound formed in intermediate of HVZ reaction.

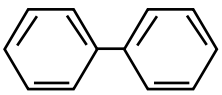
(A) 

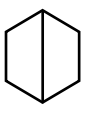
(B) 

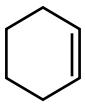
(C) 

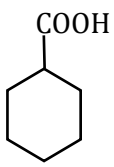
(D) R—CN

6. 

(A) 

(B) 

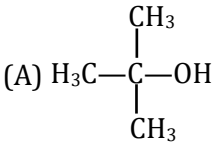
(C) 

(D) 

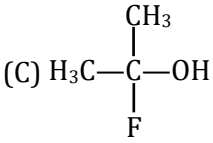
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7. $2CH_3-\overset{O}{\overset{\|}{C}}-OAg \xrightarrow{I_2} P + CO_2 + 2AgI,\ P \xrightarrow[(ii)\ H_3O^{\oplus}]{(i)\ Excess\ CH_3MgBr} (R)$ product R is

(A) 

(B) HCOOH

(C) 

(D) CH_3COOH

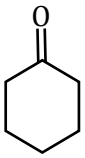
8. Which of the following step are involved in the Arndt-eistert reaction

(A) Conversion of carboxylic acid to acid chloride

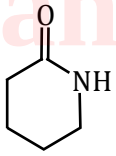
(B) Conversion of acid chloride to a diazoketone

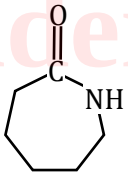
(C) Conversion of diazo ketone into ketene

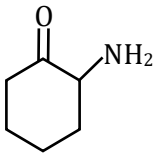
(D) All of these

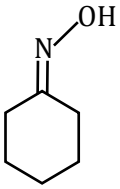
9.  $\xrightarrow[(2)\ \Delta]{(1)\ N_3H/H^{\oplus}}$ product is

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(A) 

(B) 

(C) 

(D) 

10. $CH_3-CH_2-\overset{O}{\overset{\|}{C}}-OH \xrightarrow[\Delta]{NH_3}$

product P is

(A) CH_3-CH_2-CN

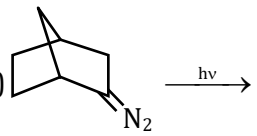
(B) $CH_3-\overset{O}{\overset{\|}{C}}-NH_2$

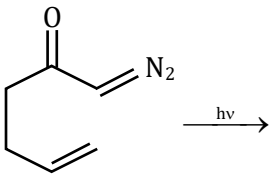
(C) $CH_3-CH_2-\overset{O}{\overset{\|}{C}}-NH_2$

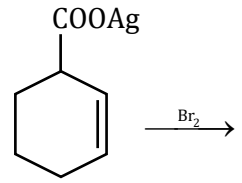
(D) $CH_3-CH_2-\overset{O}{\overset{\|}{C}}-H$

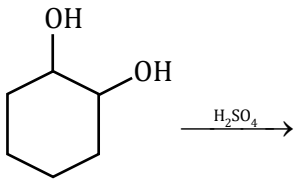
For More Join: @IITJEE_Advanced

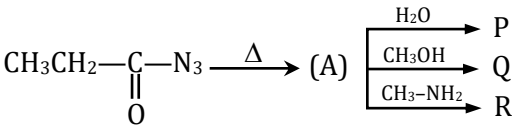
11. Which of the following compounds form carbene intermediate :

(A) 

(B) 

(C) 

(D) 

12. 

Correct product are/is

(A) P is $CH_3CH_2NH_2$

(B) $CH_3CH_2NH-\underset{\underset{O}{\|}}{C}-OCH_3$

(C) $CH_3CH_2-NH-\underset{\underset{O}{\|}}{C}-NH-CH_3$

(D) None of these

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	A	A	BC	A	A	A	D	B	C
Question	11	12								
Answer	AB	ABC								

For More Join: @IITJEE_Advanced